

RESEARCH ARTICLE

A Zero-Trust Authentication Scheme With Access Control for 6G-Enabled IoT Environments

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ABSTRACT The sixth generation (6G) is the next generation of wireless communication technology, is not limited to cellular networks but can be used to provide better services in all areas of wireless communication, including vehicles, drones, and smart homes. However, these advancements in 6G technology require further security considerations. In earlier networks, authentication schemes were designed to rely on a secure channel between the core network and access points. The 6G network is an open, distributed network that integrates multiple domains and entities and cannot guarantee secure channels in the network. The 6G network requires a new security authentication scheme that applies the zero-trust model. Therefore, this study proposes a zero-trust authentication scheme with access control for 6G-enabled Internet of Things environments. The scheme uses blockchain technology for mutual authentication in a distributed environment and lightweight attribute-based encryption to ensure dynamic access control and network efficiency. This study compares the proposed authentication method with existing methods and demonstrates that this scheme has better performance and security. To the best of our knowledge, this paper is the first to propose a specific authentication protocol with access control considering the zero-trust model in a 6G environment.

INDEX TERMS Access control, attribute-based encryption, authentication, blockchain, key agreement, sixth generation (6G), zero-trust.

I. INTRODUCTION

The sixth generation (6G) is the next generation of wireless communication technology that can support transmission speeds of about ten times faster than fifth-generation (5G) technology [1] and is expected to enter commercialization soon. In 6G communication environments, users communicate with access points (APs) or base stations via Internet of Things (IoT) devices such as smart phones, laptops, wearable devices, and sensors. In these environments, APs not only act as intermediaries in communications but can offer direct services to IoT devices, enhancing the network efficiency and service quality [2], [3]. Furthermore, 6G technology can exceed the scope of cellular networks and be integrated with

IoT networks such as smart homes, smart grids, and internet of vehicles. In conclusion, 6G technology can provide a broad range of quality services in future IoT environments.

However, the advancements of 6G technology demand heightened security considerations. A 5G network consists of user equipments (UEs), APs (base stations), and core networks (CNs), which communicate with each other using a 5G-authentication and key agreement (AKA) scheme [4]. An AP is generally a static entity, and the channel between an AP and a CN is assumed secure. Therefore, when UE communicates with an AP, the trust mechanism relies on this secure channel. However, the 6G network is more decentralized and complicated than the 5G network [5], [6]. There may not be a secure channel on the 6G network, and each element of the 6G network should establish a trust relationship through direct mutual authentication. Therefore,

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the 6G network requires a zero-trust-based authentication scheme that does not assume a secure channel. Zero-trust is a security model that follows the principle of “Never trust, always verify”. Zero-trust can provide enhanced security than perimeter security model and can meet the security requirements of distributed and wide-ranging 6G network. Furthermore, in 6G networks, the UE is not limited to smartphones but includes many IoT devices, such as wearable devices, vehicles, and drones. A zero-trust-based dynamic access control system for 6G networks must be applied to set access rights according to the UE status to use network resources efficiently [7]. The UE must request access rights from the CN after logging into the network, and the CN must be able to set access rights for the UE according to the request.

We propose a zero-trust authentication and access control scheme for 6G-enabled IoT environments to address these security challenges. The primary technologies employed in this study are discussed as follows. First, this study applies blockchain technology to ensure trustworthiness between 6G network elements without secure channels. Blockchain as a distributed ledger can guarantee decentralization and integrity of stored data and can be a security solution for 6G-enabled IoT environments. However, the blockchain incurs considerable storage costs, and IoT devices make it difficult to store all blockchain ledgers. Thus, this research employs a consortium blockchain model with the AP and CN acting as full nodes and IoT devices acting as light nodes to solve the problem. Second, the scheme adopts lightweight attribute-based encryption (ABE) [8] for fine-grained access control of the UE. In 6G-enabled IoT networks, dynamic access control must be guaranteed depending on the user status and access device, and ABE is an appropriate solution for this environment. However, traditional ABE schemes [9] require bilinear pairing and map-to-point hash operations, which are highly computationally demanding. Therefore, we employ an elliptic curve cryptosystem (ECC)-based lightweight ABE scheme suitable for IoT devices. We adopted the enhanced security model and proposed an dynamic access control scheme for 6G-enabled IoT networks. The proposed scheme provides better security compared to existing schemes and can prevent various attacks occurring in a wide range of attack surface in 6G-enabled IoT environments. The main contributions of this paper are presented below.

A. CONTRIBUTIONS

- We proposed two authentication protocols between the AP and CN and between the UE and AP. Each participant uses the blockchain to verify the other party's identity because no trusted channel exists in 6G-enabled IoT environments.
- We proposed an access control scheme between the UE and CN. We apply an ECC-based lightweight ABE for fine-grained access control [9]. An attribute set of the UE is uploaded to the blockchain after registration, and the CN can control the access rights of the UE.

- We analyzed the correctness and robustness of our scheme using BAN-logic and AVISPA tool. We also conduct informal analysis to demonstrate that the proposed scheme has resistance to various security attacks.
- We experimented the proposed scheme using network simulator 3 (NS-3). We measured the throughput and the end-to-end delay of the proposed scheme and show that the proposed scheme is practical.

B. ORGANIZATION

The paper organization is as follows. We introduce related works in Section II, explain preliminaries in Section III, and present the proposed scheme in Section IV. We analyze the security and performance of the proposed scheme in Section V and Section VI, respectively. We conclude the paper in Section VII.

II. RELATED WORKS

We introduce papers that have applied the zero-trust model in 6G networks. In 2024, Liu et al. [10] presented a zero-trust-based security architecture for 6G-enabled mobile networks. They claimed that 6G technology could integrate mobile networks and that the security architecture of the 5G network cannot satisfy the security requirements of 6G network. The proposed architecture includes data supervision based on the blockchain, dynamic situational awareness based on digital twins, and identity authentication. They also emphasized the need for a dynamic access control scheme and direct authentication protocol between network elements. Additionally, Chen et al. [7] proposed a zero-trust architecture for 6G networks and explained that 6G networks have an expanded attack surface because it integrates wireless access technology and control domains. They proposed a zero-trust architecture that harmonizes network control domains to construct an access control mechanism. Han et al. [11] designed a blockchain-based zero-trust framework for 6G radio access network (RAN). They claimed that the development of wireless communication technology inevitably increases data, creating challenges regarding data security. They used blockchain technology to ensure IoT data reliability and improved the consensus algorithm considering the large-scale application scenarios of the 6G RAN. In addition, Sedjelmaci and Tourki [12] suggested a distributed zero-trust framework for the 6G RAN. They noted that the 6G RAN must perform heavy computations on behalf of capability-reduced devices, which can cause attacks (e.g., denial of service, impersonation, and session hijacking, which are active attacks in 5G networks) and are expected to become more frequent in 6G networks. They proposed a zero-trust framework to resolve attack detection and decision-making problems in the 6G RAN. These schemes [7], [10], [11], [12] suggested a zero-trust model-based scheme for 6G security. However, they did not present specific authentication and access control protocols in 6G networks.

TABLE 1. Strengths and limitations of the related works.

Author	Year	Strengths	Limitation
Liu <i>et al.</i> [10]	2024	Proposed zero-trust-based security architecture for 6G mobile networks	Do not support specific authentication and access control schemes
Chen <i>et al.</i> [7]	2024	Proposed zero-trust-based access control mechanism integrating network control domains	
Han <i>et al.</i> [11]	2022	Presented blockchain-based-zero trust framework for large-scale application scenarios for 6G radio access networks	
Sedjelmaci and Tourki [12]	2023	Presented zero-trust framework against various attacks for 6G radio access networks	
Darman <i>et al.</i> [13]	2022	Proposed an authentication and key agreement between a user and smart industrial device for 6G-based industrial applications	Does not handle authentications between gateway and core network
Chaudhry <i>et al.</i> [14]	2023	Proposed an authentication scheme for 6G-enabled maritime transportation system against GPS jamming and ransomware attacks	Depends on a shared key issued in the registration phase
Ma <i>et al.</i> [15]	2024	Noted limitations of 5G-AKA and proposed an improved scheme for 5G/6G networks	Depends on the secure channel between AP and CN
Zhang <i>et al.</i> [16]	2021	Designed an authentication protocol for an environment with mobile gateways	Does handle authentication between the gateway and core network
Deebak and Seong [17]	2023	Classified cyber threats in drone networks and proposed RL-SMFA for 6G era	The security of the authentication protocol depends on a session random value
Roy <i>et al.</i> [18]	2024	Designed a zero-trust authentication scheme without trusted authority for vehicular networks	Assume the existence of a secure channel between RSU and TA
Cui <i>et al.</i> [19]	2020	Presented blockchain-based identity authentication scheme	Does not resolve scalability problem of a public blockchain
Xu <i>et al.</i> [20]	2020	Proposed blockchain-based redactable identity authentication scheme using the chameleon hash	Weakened integrity of the blockchain due to the chameleon hashes
Luo <i>et al.</i> [21]	2024	Designed vehicle grouping to guarantee low communication complexity	Does handle mutual authentication and the access control scheme

We review papers that have proposed an authentication and access control protocol in the 6G environment. Darman et al. [13] proposed an AKA scheme for 6G-based industrial applications. This scheme addressed mutual authentication between the user and a smart industrial device. The content server acts as a base station and relays messages during authentication. However, this scheme does not address mutual authentication between the user and base stations, which is essential for the quality of service of 6G-enabled networks. Chaudhry et al. [14] introduced an authentication scheme for 6G-enabled maritime transportation systems (MTSs). In modern MTSs, high-altitude platform systems facilitate the deployment of drone base stations. Chaudhry et al. claimed that MTSs have security problems, such as global positioning system jamming and ransomware attacks; therefore, they proposed a secure, effective authentication scheme for 6G-enabled MTSs. Furthermore, Ma et al. [15] proposed an uncrewed aerial vehicle (UAV)-assisted UE access authentication scheme for 5G and 6G networks. They stated that UAVs can act as base stations and support the network access of UE for 5G and 6G networks, noting the limitations of the 5G AKA and introducing an enhanced AKA. Their scheme assumes secure channels between the UAV-to-ground RAN and the ground RAN to the CN; however, these assumptions cannot satisfy the zero-trust security requirements in 6G networks. Zhang et al. [16] designed a drone-assisted authentication scheme for a 5G vehicle ad hoc network. In their scheme, a drone acts as a roadside unit or gateway, ensuring seamless vehicle connectivity in rural and crowded areas. Their scheme addresses the mutual key agreement between a vehicle and

drone. Deebak and Seong [17] proposed a robust lightweight secure multi-factor authentication (RL-SMFA) for 6G era. They classified cyber threats in drone networks and designed an authentication scheme utilizing ECC to resist them. Roy et al. [18] proposed a zero-trust-based authentication scheme for vehicular communication systems. They do not assume the existence of a trusted authority and proposed an direct authentication between vehicle and infrastructure. However, their scheme cannot achieve zero-trust security because they assumed the trust channel between RSU and TA. Furthermore, their scheme is vulnerable to ephemeral session random number leakage (ESL) attack.

In summary, recent research [7], [10], [11], [12] has highlighted the zero-trust model to meet the security requirements of 6G-enabled networks, presenting zero-trust-based security architectures or frameworks. These studies [7], [10], [11], [12] have mentioned the need for mutual authentication and access control but have not suggested specific protocols. In contrast, existing protocols in 6G-enabled networks [13], [14], [15], [16], [17], [18] that are not designed based on the zero-trust model or that only operate in limited areas cannot satisfy the security requirements of 6G-enabled IoT networks. Thus, this paper proposes a blockchain-based zero-trust authentication protocol for 6G networks. A secure mutual authentication protocol can be designed in situations with no trust authority using the integrity and decentralized characteristics of the blockchain. This study introduces several blockchain-based authentication schemes before presenting the proposed scheme.

Blockchain technology emerged to implement cryptocurrency but has been widely used in identity authentication

because it can guarantee data integrity. For example, Cui et al. [19] proposed blockchain-based identity authentication for multiple wireless sensor networks using public and local blockchains. The public blockchain is global and is used to authenticate the cluster head nodes, whereas the local blockchain covers a single wireless sensor network to authenticate ordinary nodes. This scheme employed the blockchain for identity authentication but did not resolve the network delay problems that occurred using the public blockchain. In addition, Xu et al. [20] created a blockchain-based redactable identity management and authentication scheme for cellular networks. In this scheme, a service provider can authenticate a user by querying the blockchain and negotiating a session key, and their scheme applies the chameleon hash technology for user revocation. Further, Luo et al. [21] introduced a blockchain-based authentication scheme called ESIA for the Internet of Vehicles, which employs a fog-computing-based environment. This scheme focused on designing vehicle grouping with low communication complexity for blockchain scalability. These schemes [19], [20], [21] contributed to demonstrating the utility of the blockchain for identity authentication. However, these schemes do not meet the security considerations of 6G-enabled IoT environments. Table 1 summarizes the strengths and limitations of these related papers.

III. PRELIMINARY

In this section, we explain main technologies used to design zero-trust authentication and access control scheme and illustrate the system model of the proposed scheme.

A. ELLIPTIC CURVE CRYPTOSYSTEM

We briefly introduce Elliptic Curve Cryptosystem (ECC) [22]. Given an ECC parameter (q, a, b, G, p) , it means an elliptic curve $E_p(a, b)$ defines $y^2 = x^3 + ax + b$ ($4a^3 + 27b^2 \neq 0$) over the finite field $\mathbb{F}_p$, and G is a subgroup of the additive group of points on $E_p(a, b)$ with large prime order q . Then, let a generator $P \in G$ and an integer $a \in \mathbb{Z}_q$, the point multiplication aP over $E_p(a, b)$ can be defined as

$\underbrace{P + P + P + \dots + P}_{a \text{ times}} = aP$. The ECC cryptosystem has following hardness problems.

- 1) **Elliptic Curve Discrete Logarithm (ECDL) problem:** When aP ($a \in \mathbb{Z}_q^*$) is given, it is hard to find a .
- 2) **Elliptic Curve Computational Diffie-Hellman (ECDH) problem:** When aP and bP ($a, b \in \mathbb{Z}_q^*$) are given, it is hard to compute abP .
- 3) **Elliptic curve decisional Diffie-Hellman (ECDDH) problem:** Even if aP, bP and kP ($a, b, c \in \mathbb{Z}_p^*$) are given, it is hard to determine whether $kP \stackrel{?}{=} abP$.

B. ADVERSARY MODEL

We adopted Dolev-Yao (DY) adversary model [23] for security analysis of the proposed scheme. Let A be a

malicious attacker, then A has the following capabilities in wireless channels.

- Adversary A can be a legitimate user in the network and has full control of the messages transmitted on wireless channels. Moreover, A can eavesdrop, delete, intercept, and modify transmitted messages [24], [25].
- Adversary A can steal UE and extract stored values via the power analysis attack [26]. Furthermore, A can try to guess a legitimate user's identity and password [27], [28].
- Adversary A can attempt passive and active attacks using information obtained from transmitted messages and extracted values. These attacks include session key disclosure, impersonation, replay, man-in-the-middle (MITM) attacks, and others [29], [30], [31].

C. LIGHTWEIGHT ATTRIBUTE-BASED ENCRYPTION

Attribute-based encryption (ABE) is first proposed in [9], and it can provide fine-grained access control by attribute keys and access structures. However, the traditional ABE used computationally heavy operations such as bilinear pairing and map-to-point hash, and it is not suitable for resource-limited devices. In 2014, Yao et al. proposed a lightweight ABE for IoT using ECC [8]. The lightweight ABE can be operated based on an elliptic curve with parameters (q, a, b, G, p) . Their scheme utilizes the same access structure as the traditional ABE [9], and the detailed algorithms are as follows.

- **Setup:** For attribute space $\mathbb{A} = \{1, 2, \dots, n\}$ and each attribute ($i \in \mathbb{A}$), choose a secret key $s \in \mathbb{Z}_q^*$ and a random number $s_i \in \mathbb{Z}_q^*$, and computes public keys and $PK = s.P$ $P_i = s_i.P$. Then, the public parameters are $\{PK, P_1, P_2, \dots, P_n\}$
- **Encryption** (M, ω): The message M is encrypted based on elliptic curve integrated encryption scheme (ECIES). Let M is encrypted through attribute ω . Then, choose a random number $r \in \mathbb{Z}_q^*$ and compute $C' = r.PK = (K_x, K_y)$ ($C' \neq 0$) and $C_i = r.P_i$ ($i \in \omega$). Then, K_x is the symmetric encryption key and K_y is the integrity key for message M . Compute $C = \text{sym}(M, K_x)$ and $H_M = H(M, K_y)$. The ciphertext $\{M\}_\omega = (\omega, C, H_M, C_i$ ($i \in \omega$))
- **Key Generation** (Γ, s): A decryption key which satisfies the access tree $\Gamma(\omega)$ is necessary to decrypt $\{M\}_\omega$. For each node u of Γ , a polynomial $q_u(x)$ with order $(d_u - 1)$ should be defined. These polynomials can be determined in top-down manner with $q_\gamma(0) = s$ (γ is a root node of Γ). Then, the decryption key $D = \frac{q_u(0)}{s_i}$ ($i \in \omega$).
- **Decryption** ($\{M\}_\omega, D$): The decryption is the same as the traditional ABE. For each leaf node u , compute as follows.

$$\begin{aligned} \text{Decrypt}(\{M\}_\omega, D, u) &= D_u.C_i = q_u(0) \cdot \frac{1}{s_i} \cdot r.P_i \\ &= q_u(0) \cdot r.P \end{aligned}$$

For each non-leaf node u , its child node v , and compute as follows.

$$\begin{aligned} & Decrypt(\{M\}_\omega, D, v) \\ &= \sum_{v \in \omega_u} \Delta_{i, w'_u}(0).q_v(0).r.P \\ &= \sum_{v \in \omega_u} \Delta_{i, w'_u}(0).q_{parent(v)}(index(v)).r.P \\ &= q_u(0).r.G \end{aligned}$$

where ω_u is a set of v and $i = index(v)$, $\omega'_u = \{index(v), v \in \omega_u\}$. Repeat this process to obtain the decryption key $Decrypt(\{M\}_\omega, D, \gamma) = q_\gamma(0).r.P = s.k.P = (K_x, K_y)$. Then, K_x can be used as the decryption key and K_y can be used as the message integrity key.

D. SYSTEM MODEL

The proposed model comprises the UE, AP, CN, and the blockchain. The UE is a terminal through which users can access the network. A smartphone, vehicle, or wearable device could be employed as the UE. The UE has a low computational capacity, and internal data can be leaked through power analysis attacks. The AP relays the connection between the UE and CN and directly provides services to the UE. In addition, the AP includes mobile devices, such as drones; thus, a secure channel between the CN and AP cannot be guaranteed. The CN communicates with the AP and sets the access control permissions for users. The CN is assumed to be trusted but does not have a secure channel established with other entities. The blockchain stores the identity information for the UE, APs, and CNs and the access rights of the UE. The UE and mobile APs can act as light nodes, and the CN and static APs can act as heavy nodes to store the blockchain. Fig. 1 presents the proposed system model.

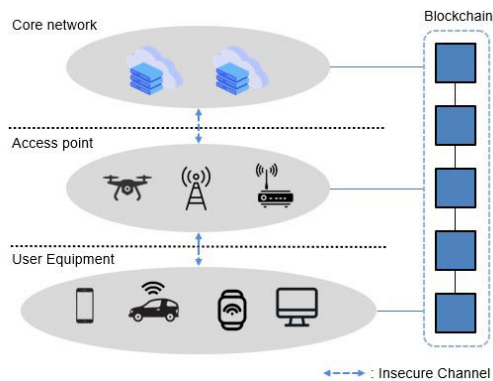


FIGURE 1. The proposed system model.

IV. PROPOSED SCHEME

The proposed scheme consists of initialization, registration, authentication, and access control phases. Before we introduce the proposed scheme, we explain the notations and their

meaning used in our scheme, and provide a brief description of our scheme.

TABLE 2. Notations and their meanings.

Notation	Description
CN_i	i -th core network
AP_j	j -th access point
UE_k	k -th user equipment
$\mathbb{A}$	attribute universe
$\{a_m\}_{m=1}^n$	attribute keys
P_i	public key of CN_i
T_1	timestamp
l	a fuzzy verifier
Req_k	a access control request message of UE_k
$\{a_{l_k}\}$	an attribute key set of UE_k
$\{Req_k\}_{\{a_{l_k}\}}$	Req_k is encrypted with attribute keys $\{a_{l_k}\}$
SK_{ij}	Session key between CN_i and AP_j

In the initialization phase, KGC generates a public/secret key pair and selects a hash operation for the network. After that, in the registration phase, KGC registers CN , AP , and UE and issues secret parameters to allow them join the network. Then, AP - CN and UE - AP authenticate each other to agree on a session key for further communications. Using the session keys, UE can request access right to CN . In this phase, AP acts as a middleman between UE and CN . Fig. 2 represents the flowchart of the proposed scheme.

A. INITIALIZATION

In the initialization phase, KGC chooses a large prime q and set an elliptic curve $E_q(a, b) = y^2 = x^3 + ax + b$ with order p . Then, KGC chooses a fuzzy verifier $l : 2^4 \leq l \leq 2^8$, a generator P , and hash function $h(\cdot)$. It generates the attribute universe $\mathbb{A} = \{A_1, A_2, \dots, A_n\}$ and the corresponding attribute keys $\{a_1, a_2, \dots, a_n\}$, and computes $P_1 = a_1.P, \dots, P_n = a_n.P$. It also publishes system public parameters $(q, p, E_q(a, b), P, \{P_m\}_{m=1}^n h(\cdot))$.

B. REGISTRATION

1) CN REGISTRATION

First, KGC selects a unique identity ID_i and a random $s_i \in \mathbb{Z}_p^*$ and computes $P_i = s_i.P$, generating a blockchain wallet with the address P_i . Afterward, KGC publishes (ID_i, P_i) and sends $(s_i, \{a_m\}_{m=1}^n)$ to CN_i , and CN_i stores them securely.

2) AP REGISTRATION

The AP registration phase is conducted similarly to the CN registration phase. KGC sends s_j to AP_j , and AP_j stores s_j securely.

3) UE REGISTRATION

UE_k chooses an identity ID_k and password PW_k , access right AR_k , and a random number $e_k \in \mathbb{Z}_p^*$. Next, it computes $HPW_k = h(PW_k || e_k)$ and sends (ID_k, HPW_k) to KGC . Then, KGC checks whether ID_k is registered, and if not, it generates $s_k \in \mathbb{Z}_p^*$, computes $P_k = s_k.P$, and generates a blockchain wallet with address P_k . Then, KGC publishes (ID_k, P_k) , uploads access right AR_k of UE_k to the smart

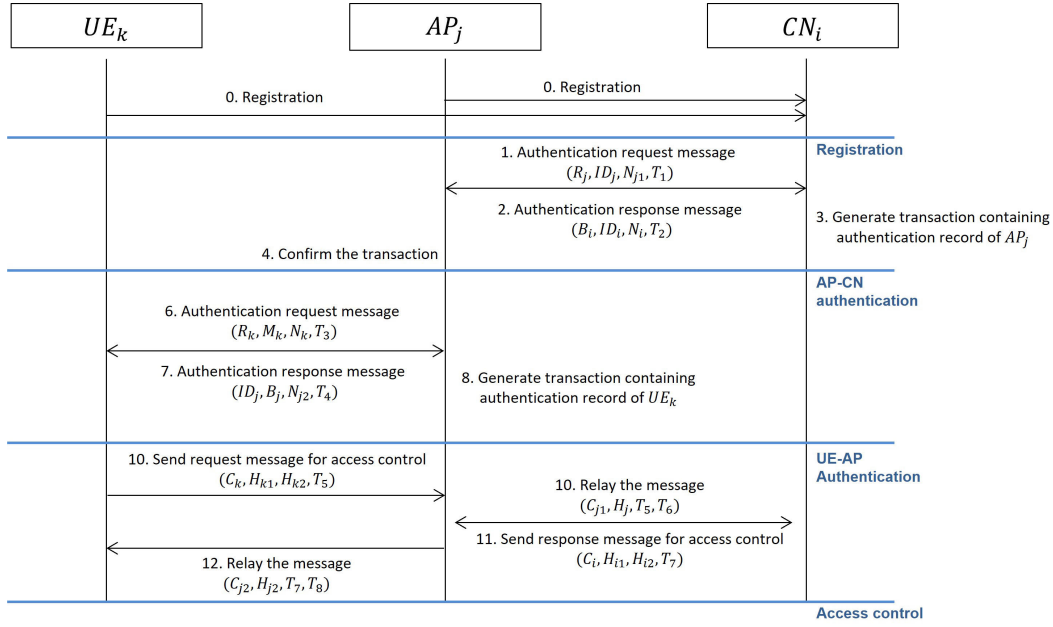


FIGURE 2. The flowchart of the proposed scheme.

contract on the blockchain, and sends $(s_k, \{a_{l_k}\}_{l_k \in AR_k}, l)$ to UE_k . UE_k computes $HID_k = h(ID_k || PW_k)$, $A_k = e_k \oplus HID_k$, $B_k = (s_k || \{a_{l_k}\}_{l_k \in AR_k}) \oplus HPW_k$, and $Auth_k = h(e_k || s_k || \{a_{l_k}\}_{l_k \in AR_k} || ID_k || PW_k) \pmod{I}$ and stores $(A_k, B_k, Auth_k, l)$ in the secure memory.

C. AUTHENTICATION

1) AP-CN AUTHENTICATION

AP_j generates $r_j \in Z_p^*$ and a timestamp T_1 , computes $R_j = r_j.P$, $R_{ij} = r_j.P_i$, $P_{ij} = s_j.P_i$, and $N_{j1} = h(ID_j || R_{ij} || P_{ij} || T_1)$, and transmits (R_j, ID_j, N_{j1}, T_1) to CN_i . CN_i checks the validity of T_1 , computes $R_{ij} = s_i.R_j$ and $P_{ij} = s_j.P_i$, retrieves blockchain address P_j using ID_j , and checks whether $N_{j1} \stackrel{?}{=} h(ID_j || R_{ij} || P_{ij} || T_1)$. If it is equal, CN_i generates $r_i \in Z_p^*$ and T_2 , computes $B_i = r_i.P$, $B_{ij} = r_i.R_j$, $SK_{ij} = h(R_{ij} || B_{ij} || P_{ij} || ID_j || ID_i)$, and $N_i = h(B_{ij} || P_{ij} || R_{ij} || SK_{ij})$. After that, CN_i generates a transaction containing authentication record and transmits (B_i, ID_i, N_i, T_2) to AP_j . AP_j checks the validity of T_2 , computes $B_{ij} = r_j.B_i$ and $SK_{ij} = h(R_{ij} || B_{ij} || P_{ij} || ID_j || ID_i)$, and verifies $N_i \stackrel{?}{=} h(B_{ij} || P_{ij} || R_{ij} || SK_{ij})$. If it is equal, AP_j authenticates CN_i and confirms the transaction generated by CN_i .

2) UE-AP AUTHENTICATION

A user inputs ID_k and PW_k to UE_k . Then, UE_k computes $HID_k = h(ID_k || PW_k)$, $e_k = A_k \oplus HID_k$, $HPW_k = h(PW_k || e_k)$, and $(s_k || \{a_{l_k}\}) = B_k \oplus HPW_k$ and checks whether $Auth_k \stackrel{?}{=} h(e_k || s_k || \{a_{l_k}\} || ID_k || PW_k) \pmod{I}$. If it is equal, the user successfully logged in to UE_k . After that, UE_k generates $r_k \in Z_p^*$ and T_3 , computes

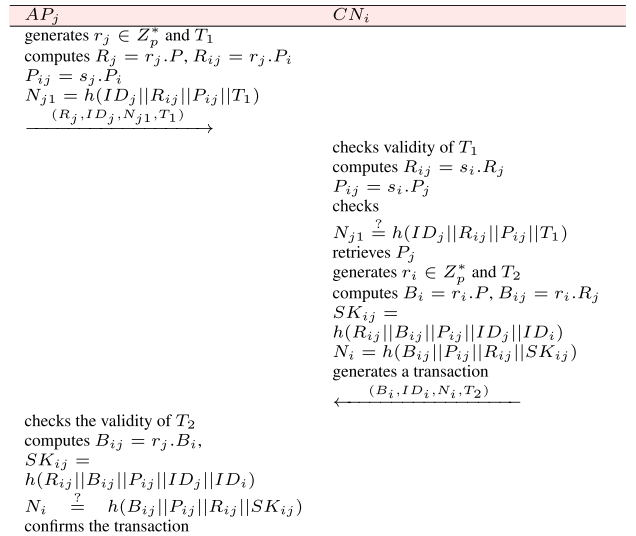


FIGURE 3. Proposed AP-CN authentication phase.

$R_k = r_k.P$, $R_{kj} = r_k.P_j$, $P_{kj} = s_k.P_j$, $M_k = (ID_k || P_k) \oplus h(R_{kj})$, and $N_k = h(ID_k || R_{kj} || P_{kj} || T_3)$. UE_k transmits (R_k, M_k, N_k, T_3) to AP_j . AP_j checks the validity of T_3 , computes $R_{kj} = s_j.R_k$, $(ID_k || P_k) = M_k \oplus h(R_{kj})$, and retrieves blockchain address P_k . Then, AP_j computes $P_{kj} = s_j.P_k$ and checks $N_k \stackrel{?}{=} h(ID_k || R_{kj} || P_{kj} || T_3)$. If it is equal, AP_j generates $c_j \in Z_p^*$ and T_4 and computes $B_j = c_j.P$, $B_{kj} = c_j.R_k$, $SK_{kj} = h(R_{kj} || B_{kj} || P_{kj} || ID_j || ID_k)$, and $N_{j2} = h(B_{kj} || P_{kj} || R_{kj} || SK_{kj})$. AP_j generates a transaction containing authentication record and transmits (ID_j, B_j, N_{j2}, T_4) to UE_k . UE_k checks the validity of T_4 and computes

$r_k.B_j = B_{kj}$ and $SK_{kj} = h(R_{kj}||B_{kj}||P_{kj}||ID_j||ID_k)$ and checks $N_{j2} \stackrel{?}{=} h(B_{kj}||P_{kj}||R_{kj}||SK_{kj})$. If it is equal, UE_k confirm the transaction.

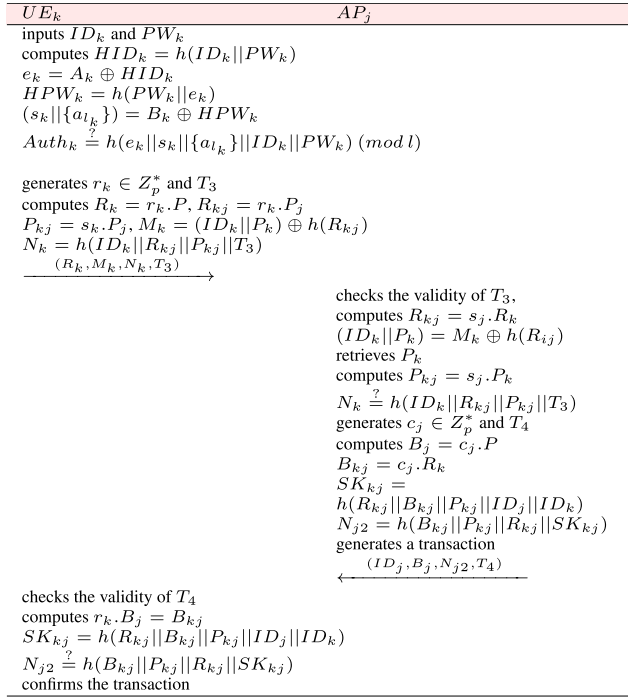


FIGURE 4. Proposed UE-AP authentication phase.

D. ACCESS CONTROL

For UE_k to request a new access right, UE_k generates a request Req_k and T_5 , encrypts $C_k = (ID_k||\{Req_k\}_{a_{l_k}}) \oplus SK_{kj}$, computes $H_{k1} = h(ID_k||\{Req_k\}_{a_{l_k}}||T_5)$ and $H_{k2} = h(ID_k||\{Req_k\}_{a_{l_k}}||SK_{kj}||T_5)$, and sends $(C_k, H_{k1}, H_{k2}, T_5)$ to AP_j . AP_j checks the validity of T_5 , decrypts $(ID_k||\{Req_k\}_{a_{l_k}}||T_5) = C_k \oplus SK_{kj}$, checks $H_{k1} \stackrel{?}{=} h(ID_k||\{Req_k\}_{a_{l_k}}||T_5)$. If it is equal, AP_j generates T_6 , encrypts $C_{j1} = (ID_k||\{Req_k\}_{a_{l_k}}||T_6) \oplus SK_{ij}$, computes $H_j = h(H_{k2}||T_6)$, and transmits (C_{j1}, H_j, T_5, T_6) to CN_i . CN_i checks the validity of T_6 , decrypts $(ID_k||\{Req_k\}_{a_{l_k}}||T_6) = C_{j1} \oplus SK_{ij}$, and verifies $H_{k2} \stackrel{?}{=} h(ID_k||\{Req_k\}_{a_{l_k}}||SK_{kj}||T_5)$. If it is equal, CN_i checks AR_k from blockchain and decrypts Req_k . Then, CN_i generates a response message Res_k and T_7 , computes $C_i = (ID_i||Res_k) \oplus SK_{kj} \oplus h(s_i.P_k)$, $H_{i1} = h(ID_i||Res_k) \oplus h(s_i.P_k||T_7)$, and $H_{i2} = h(ID_i||Res_k||T_7)$. Then, CN_i sends $(C_i, H_{i1}, H_{i2}, T_7)$ to AP_j . AP_j checks the validity of T_7 , decrypts $(ID_i||Res_k) = C_i \oplus SK_{kj} \oplus h(s_i.P_k)$ and checks $H_{i1} \stackrel{?}{=} h(ID_i||Res_k) \oplus h(s_i.P_k||T_7)$. If it is equal, AP_j generates T_8 , encrypts $C_{j2} = (ID_i||Res_k) \oplus h(s_i.P_k) \oplus SK_{kj}$, computes $H_{j2} = h(H_{i2}||T_8)$, and sends $(C_{j2}, H_{j2}, T_7, T_8)$ to UE_k . UE_k checks the validity of T_8 , decrypts $(ID_i||Res_k) = C_{j2} \oplus h(s_i.P_k) \oplus SK_{kj}$, computes $H_{i2} = h(ID_i||Res_k||T_7)$, and checks $H_{j2} \stackrel{?}{=} h(H_{i2}||T_8)$. Then, the proposed access control phase is finished.

V. SECURITY ANALYSIS

In this section, we analyze the proposed scheme against various attacks and through Burrows-Abadi-Needham (BAN) logic [32], Real-or-Random (RoR) model [33], and Automated Validation of Internet Security Protocols and Applications (AVISPA) tool [34].

A. INFORMAL ANALYSIS

We demonstrate that the proposed authentication and access control schemes have resistance to malicious attacks and can provide various security features. We assume that an adversary A can attempt attacks using their capabilities we described in Section III-B.

1) REPLAY AND MITM ATTACKS

A can reuse the transmitted message or act as a middleman during communications. In the proposed scheme, each message contains a timestamp and a hash value calculated using a Diffie-Hellman symmetric key $(P_{ij}, B_{ij}, R_{kj}, P_{kj})$ or a session key (SK_{kj}, SK_{ij}) . Therefore, if A reuse the message, the timestamp will expire and the message will be considered invalid. Furthermore, A cannot modify the message because every message include the message digest and it is hashed using a symmetric key. Therefore, A fails to perform replay and MITM attacks.

2) DEVICE STOLEN ATTACK

A can steal UE_k and attempt logging in to the UE_k for participating in the network. According to the assumption, A can extract the stored value in $UE_i: (A_k, B_k, Auth_k)$ through power analysis attack. A must successfully guess ID_k and PW_k to obtain secret keys of UE_k . However, $Auth_k$ in UE_k is masked using a fuzzy verifier l . Even if A succeeds in logging in to UE_k , the probability that the guessed ID_k and PW_k are correct is $\frac{1}{10^{12}} \approx \frac{1}{2^{32}}$, which can be neglected.

3) ACCESS POINT CAPTURE ATTACK

In 6G-enabled IoT environments, drones and mobile devices can act as access points and A can attempt capture attacks against these devices. Then, A can obtain the stored value s_j , which is a secret key of AP_j . Generally, if an AP is compromised, then the AP is added to revocation list and cannot communicate with other entities in the network. Furthermore, A cannot obtain any information about other APs and CNs using s_j , and A cannot effect to the network other than compromising a single AP. Therefore, the proposed scheme has resistance to the access point capture attack.

4) IMPERSONATION ATTACK

A can disguise as a network participant and communicate with other entities in the proposed authentication and access control phases. In the AP-CN authentication phase, A cannot generate P_{ij} without knowing secret key s_i or s_j , and in the UE-AP authentication phase, A cannot generate P_{kj} without knowing s_j or s_k . In the access control phase, A cannot obtain

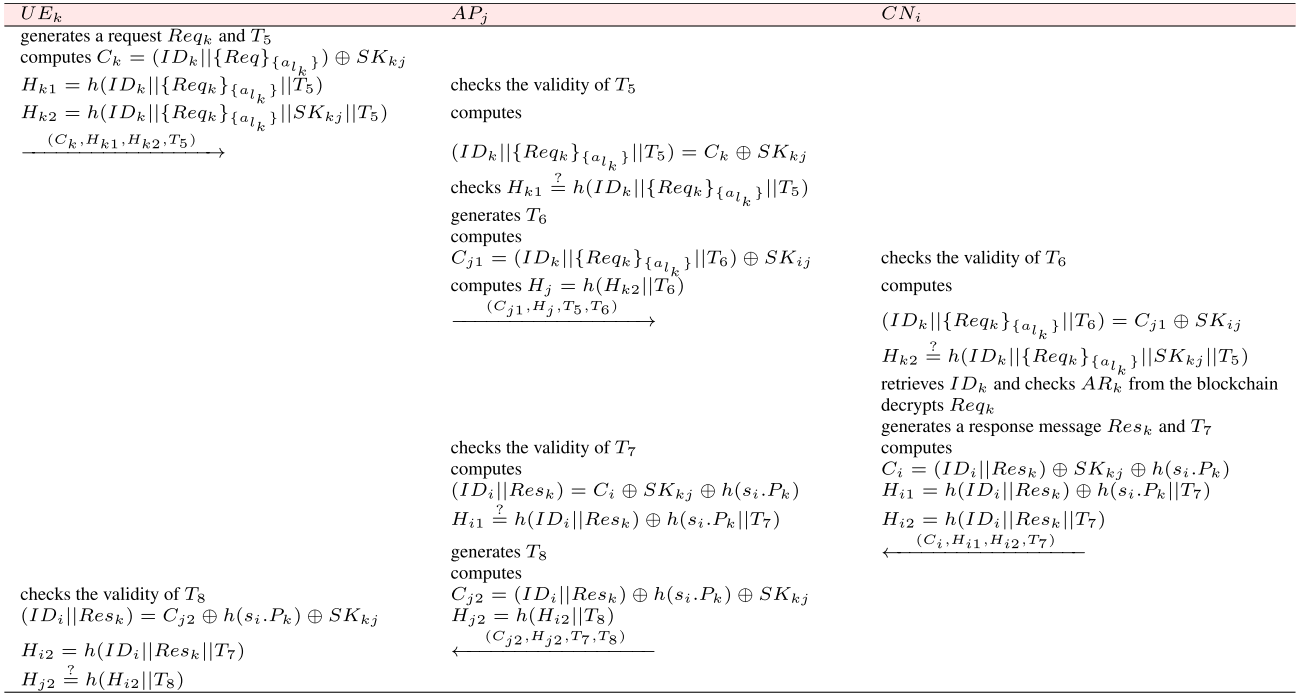


FIGURE 5. Proposed access control scheme.

session keys SK_{kj} and SK_{ij} . Therefore, A cannot impersonate to legitimate participant.

5) SESSION KEY DISCLOSURE ATTACK

A can try to calculate the session key using the transmitted messages in public channels and obtained values. A must guess identities and obtain symmetric keys to calculate SK_{ij} and SK_{kj} , but A cannot obtain any of these values. Therefore, the proposed scheme has resistance to session key disclosure attack.

6) EPHEMERAL SESSION RANDOM NUMBER LEAKAGE (ESL) ATTACK

When A obtains session random numbers r_j and r_i in the AP-CN authentication phase and r_k and c_j in the UE-AP authentication phase. Afterwards, A can try to calculate a session key of these sessions. However, A cannot calculate SK_{ij} without knowing P_{ij} and cannot calculate SK_{kj} without knowing P_{kj} because these values are calculated using long-term secret keys of each entity. Therefore, the proposed scheme has resistance to ESL attack.

7) ANONYMITY AND UNTRACEABILITY

If an identity of a specific UE is transmitted in a public channel or if A can trace a specific UE using transmitted messages in the public channel, it may infringe the privacy of the UE. In the proposed scheme, the transmitted messages does not contain a static value or an identity to track an UE. Therefore, the proposed scheme support anonymity and untraceability.

8) ACCESS CONTROL

In the proposed access control phase, UE_k generates a request message Req_k and encrypts it using $\{a_{l_k}\}$, which is a set of attribute keys of UE_k issued in the registration phase. Then, CN_i can check the access right AR_k of UE_k through the blockchain, decrypt Req_k , generate a response message Res_k , and sends it to UE_k . Therefore, the proposed scheme provides access control.

B. FORMAL SECURITY ANALYSIS USING BAN LOGIC

We analyze the proposed authentication protocols using BAN-logic [32] to show that our protocols are correct. We perform the BAN logic analysis for the AP-CN authentication because the BAN logic analysis of UE-AP authentication can be performed in a same way.

TABLE 3. Notations and meanings.

Notation	Meaning
e_1, e_2	Communication entities
s_1, s_2	Statements
$e_1 \models s_1$	e_1 believes s_1
$e_1 \sim s_1$	e_1 once said s_1
$e_1 \Rightarrow s_1$	e_1 controls s_1
$e_1 \triangleleft s_1$	e_1 receives s_1
$\#s_1$	s_1 is fresh
$\{s_1\}_K$	s_1 is encrypted with K
$e_1 \xleftrightarrow{K} e_2$	e_1 and e_2 have shared key K

1. Message meaning rule (MMR):

$$\frac{e_1 \models e_1 \xleftrightarrow{K} e_2, \quad e_1 \triangleleft \{s_1\}_K}{e_1 \models e_2 \sim s_1}$$

2. Nonce verification rule (NVR):

$$\frac{e_1 \equiv \#(s_1), \quad e_1 \equiv e_2 \mid \sim s_1}{e_1 \equiv e_2 \equiv s_1}$$

3. Jurisdiction rule (JR):

$$\frac{e_1 \equiv e_2 \implies s_1, \quad e_1 \equiv e_2 \equiv s_1}{e_1 \mid \equiv s_1}$$

4. Belief rule (BR):

$$\frac{e_1 \mid \equiv (s_1, s_2)}{e_1 \mid \equiv s_1}$$

5. Freshness rule (FR):

$$\frac{e_1 \mid \equiv \#(s_1)}{e_1 \mid \equiv \#(s_1, s_2)}$$

1) GOALS

The goals of our scheme in BAN logic analysis are as follows.

- Goal 1:** $AP_j \mid \equiv AP_j \xleftrightarrow{SK_{ij}} CN_i$
Goal 2: $AP_j \mid \equiv CN_i \mid \equiv AP_j \xleftrightarrow{SK_{ij}} CN_i$
Goal 3: $CN_i \mid \equiv AP_j \xleftrightarrow{SK_{ij}} CN_i$
Goal 4: $CN_i \mid \equiv AP_j \mid \equiv AP_j \xleftrightarrow{SK_{ij}} CN_i$

2) ASSUMPTIONS

The BAN logic assumptions of our protocol are as follows.

- $A_1:$ $CN_i \mid \equiv \#(T_1)$
 $A_2:$ $AP_j \mid \equiv \#(T_2)$
 $A_3:$ $CN_i \mid \equiv AP_j \Rightarrow (AP_j \xleftrightarrow{SK_{ij}} CN_i)$
 $A_4:$ $AP_j \mid \equiv CN_i \Rightarrow (AP_j \xleftrightarrow{SK_{ij}} CN_i)$
 $A_5:$ $CN_i \mid \equiv AP_j \xleftrightarrow{P_{ij}} CN_i$
 $A_6:$ $AP_j \mid \equiv AP_j \xleftrightarrow{P_{ij}} CN_i$

3) IDEALIZED FORMS

The idealized forms of our protocol are described as follows.

- $M_1:$ $AP_j \rightarrow CN_i : [R_j, ID_j, \{R_{ij}\}_{P_{ij}}, T_1]$
 $M_2:$ $CN_i \rightarrow AP_j : [B_i, ID_i, \{B_{ij}\}_{P_{ij}}, T_2]$

4) BAN LOGIC IMPLEMENTATION

The BAN logic implementation of our scheme can be performed as follows.

- S1:** CN_i receives M_1 .

$$CN_i \triangleleft [R_j, ID_j, \{R_{ij}\}_{P_{ij}}, T_1]$$

- S2:** CN_i can decrypt the message and obtain R_{ij} . Then, CN_i recognizes that the message is from AP_j .

$$S_2: CN_i \mid \equiv AP_j \mid \sim [R_j, ID_j, R_{ij}, T_1]$$

- S3:** The FR can be applied to the message using A_1 .

$$S_3: CN_i \mid \equiv \# [R_j, ID_j, R_{ij}, T_1]$$

- S4:** The NVR can be applied using S_2 and S_3 .

$$S_4: CN_i \mid \equiv AP_j \mid \equiv [R_j, ID_j, R_{ij}, T_1]$$

- S5:** The BR can be applied using S_4 .

$$S_5: CN_i \mid \equiv AP_j \mid \equiv [R_{ij}]$$

- S6:** AP_j receives M_2 .

$$S_6: AP_j \triangleleft [B_i, ID_i, \{B_{ij}\}_{P_{ij}}, T_2]$$

- S7:** AP_j can decrypt the message and obtain B_{ij} . Then, AP_j recognizes that the message is from CN_i .

$$S_7: AP_j \mid \equiv CN_i \mid \sim [B_i, ID_i, B_{ij}, T_2]$$

- S8:** The FR can be applied using A_2 .

$$S_8: AP_j \mid \equiv \# [B_i, ID_i, B_{ij}, T_2]$$

- S9:** The NVR can be applied using S_7 and S_8 .

$$S_9: AP_j \mid \equiv CN_i \mid \equiv [B_i, ID_i, B_{ij}, T_2]$$

- S10:** The BR can be applied using S_9 .

$$S_{10}: AP_j \mid \equiv CN_i \mid \equiv [B_{ij}]$$

- S11:** CN_i and AP_j can calculate the same session key $SK_{ij} = h(R_{ij} || B_{ij} || P_{ij} || ID_j || ID_i)$.

$$S_{11}: CN_i \mid \equiv AP_j \mid \equiv (AP_j \xleftrightarrow{SK_{ij}} CN_i) \quad (\text{Goal 4})$$

and,

$$S_{12}: AP_j \mid \equiv CN_i \mid \equiv (AP_j \xleftrightarrow{SK_{ij}} CN_i) \quad (\text{Goal 2})$$

- Step 12:** The JR can be respectively applied to S_{11} and A_3 , and S_{12} and A_4 .

$$S_{13}: CN_i \mid \equiv (AP_j \xleftrightarrow{SK_{ij}} CN_i) \quad (\text{Goal 3})$$

and,

$$S_{14}: AP_j \mid \equiv (AP_j \xleftrightarrow{SK_{ij}} CN_i) \quad (\text{Goal 1})$$

We have analyzed the proposed scheme using BAN logic analysis. According to the analysis results, it can be verified that AP and CN have successfully reached mutual session key agreement. Since authentication between UE and AP is also performed according to the same procedure as authentication between AP and CN, BAN logic analysis can be performed in the same way. Therefore, the proposed authentication scheme is correct and ensures mutual authentication.

C. FORMAL SECURITY ANALYSIS USING RoR MODEL

We use the Real-or-Random (RoR) model [33] to prove the session key security of the proposed authentication protocols. Under the RoR model, and adversary attempt various queries to obtain the session key, and we calculate the probability that

the adversary will succeed in distinguishing a random number from a session key. We denote p^t be the t -th instance of an execution participant. For instance, $p_{AP_j}^1$, $p_{CN_i}^2$, and $p_{UE_k}^3$ are instances of AP_j , CN_i , and UE_k respectively. A can perform the following queries explained in Table 4.

TABLE 4. Queries and description.

Query	Description
$Execute(p_{AP_j}^1, p_{CN_i}^2, p_{UE_k}^3)$	It represents a passive attack (i.g. eavesdropping) performed by A . A can get the transmitted message over public channels.
$CorruptUE(p_{UE_k}^3)$	It represents the UE stolen attack. A corrupt an UE and can extract the stored values in the UE.
$Send(p^t, M)$	It represents an active attack performed by A . A generates a message and sends it to other participant, and can receive the response.
$Hash(p^t)$	It represents a one-way hash operation performed by A .
$Test(p^t)$	This query is to verify the semantic security of the session key in an authentication protocol. When A performs this query, an unbiased coin c , with heads representing 1 and tails representing 0, is flipped. Then, A receives the session key when $c = 1$ and receives a random number when $c = 0$. Otherwise, A receives <i>NULL</i> .

We prove the semantic security of the session key agreed in the proposed authentication protocol through RoR model.

Theorem 1: We denote Adv_A is the advantage function when A tries to distinguish the session key and a random number. Then, the following inequality holds:

$$Adv_A \leq \frac{q_{Hash}^2}{|Hash|} + \frac{q_{send}}{|D_{ID}||D_{PW}|}. \quad (1)$$

In the equation, $|Hash|$ represents the range space of the *Hash* output, and q_{Hash} and q_{send} mean the number of performed *Hash* and *Send* queries.

proof: A can perform four games G_i ($i = 1, 2, 3$). If A can success to distinguish the shared key and a random number, A wins the game.

G_0 : Before starting the game, assume that A performs *Test* query in a situation where there is no information. Then, the probability that A guesses correctly whether the output is the session key or a random value is exactly $1/2$. Then, we can obtain the following Equation (2) by the definition of semantic security.

$$Adv_A = |2Adv_A^{G_0} - 1|. \quad (2)$$

G_1 : A can conduct *Execute* query in G_1 . A eavesdrops public channels and can obtain transmitted messages (R_j, ID_j, N_{j1}, T_1) , (B_i, ID_i, N_i, T_2) , (R_k, M_k, N_k, T_3) , and (ID_j, B_j, N_{j2}, T_4) . In this game, A cannot obtain any information about session keys SK_{ij} and SK_{kj} using these values. Therefore, the advantage function of A is the same when G_0 ends. Accordingly, Equation (3) holds.

$$Adv_A^{G_1} = Adv_A^{G_0}. \quad (3)$$

G_2 : A can perform *Hash* queries using the obtained values from G_1 in this game. However, A cannot obtain any

meaningful value to calculate SK_{ij} or SK_{kj} because all the values for calculating the session keys are masked. In order for A to distinguish the session key, it has no choice but to generate a hash collision and obtain a hash output identical to the output of the session key. The advantage function of A can be induced using the birthday paradox [38].

$$|Adv_A^{G_2} - Adv_A^{G_1}| \leq \frac{q_{Hash}^2}{2|Hash|} \quad (4)$$

G_3 : In this game, A can perform *CorruptUE* and *Send* queries to obtain SK_{kj} . A can try to disguise as UE_k , generate a legitimate message, and transmit it to AP_j . Then, A should be able to generate (R_k, M_k, N_k, T_3) successfully. A can generate R_k and T_3 , but cannot generate N_k without knowing P_{kj} , which is a symmetric key between UE_k and AP_j . Therefore, A must be able to guess ID_k and PW_k simultaneously to win the game. Let D_{ID} and D_{PW} be an identity dictionary and a password dictionary, respectively. Then, the advantage function of A according to these probabilities is as following Equation (5).

$$|Adv_A^{G_3} - Adv_A^{G_2}| \leq \frac{q_{send}}{|D_{ID}||D_{PW}|}. \quad (5)$$

Still, A cannot obtain meaningful value to distinguish session key and a random number correctly. Consequently, the following Equation (6) holds:

$$Adv_A^{G_3} = \frac{1}{2} \quad (6)$$

Using Equations (2), (3), and (6), we can get the following Equation (7):

$$\begin{aligned} \frac{1}{2}Adv_A &= |Adv_A^{G_0} - \frac{1}{2}| \\ &= |Adv_A^{G_1} - Adv_A^{G_3}| \end{aligned} \quad (7)$$

Applying the triangular inequality to the Equation (7), then

$$\begin{aligned} Adv_A &= 2|Adv_A^{G_1} - Adv_A^{G_3}| \\ &\leq 2|Adv_A^{G_1} - Adv_A^{G_2}| + 2|Adv_A^{G_2} - Adv_A^{G_3}| \\ &\leq \frac{q_H^2}{|Hash|} + \frac{2q_{send}}{|D_{ID}||D_{PW}|} \end{aligned} \quad (8)$$

The advantage function of A is negligible and the semantic security of the proposed authentication protocol is guaranteed.

D. FORMAL SECURITY VERIFICATION: AVISPA SIMULATION

We simulated the proposed scheme using “Automated Validation of Internet Security Protocols and Applications (AVISPA)” simulation tool to prove that the proposed scheme is secure against replay and MITM attacks. AVISPA tool utilizes High-Level Protocol Specification Language (HLPSSL) [35] to implement an authentication protocol, and it can be simulated using following four back-ends models: 1) “On-the-Fly Model Checker (OFMC)” [36], 2) “Constraint Logic-based Attack Searcher (CL-AtSe)” [37],

3) “Tree Automata based on Automatic Approximations for Analysis of Security Protocol (TA4SP)”, and 4) “SAT-based Model Checker (SATMC)”. The HLPSSL code can be input to one the four models, the input is converted to “Intermediate Format (IF)”, and generates the output “Output Format (OF)”. We can say that an authentication is secure if the simulation results of an authentication protocol is SAFE under OFMC and CL-AtSe models. Fig. 6 shows the session, environment, and goal of the proposed scheme implemented through HLPSSL code. We simulated both AP-CN and UE-AP authentications and analyze the results. The simulation results are summarized in Fig 7. As shown in 7, the proposed authentication protocols are SAFE under the two models, and we can say that the proposed scheme is secure against replay and MITM attacks.

```

role session(UE,AP,CN:agent, SKUA,SKUC,SKAC:symmetric_key,H:hash_func)
def=
local SN1, SN2, SN3, RV1, RV2, RV3: channel(dy)
composition
equipment(UE,AP,CN, SKUA,SKUC,SKAC, H, SN1, RV1)
/W access(UE,AP,CN, SKUA,SKUC,SKAC, H, SN1, RV1)
/W core(UE,AP,CN, SKUA,SKUC,SKAC, H, SN1, RV1)
end role

role environment()
def=
const ue,ap,cn: agent,
skua,skuc,skac: symmetric_key,
h,mul: hash_func,
id,i,j,k,p,q,r,s,t: text,
ap_cn_1,ap_cn_2,ap_cn_3,ue_ap_1,ue_ap_2,ue_ap_3,ue_ap_4: protocol_id

intruder_knowledge = {id,i,j,k,p,q,r,s,t, h, mul}
composition
session(ue,ap,cn,skua,skuc,skac,h)
/W session(i,ap,cn,skua,skuc,skac,h)
/W session(ue,i,cn,skua,skuc,skac,h)
/W session(ue,ap,i,skua,skuc,skac,h)

end role

goal
authentication_on ap_cn_1,ap_cn_2,ap_cn_3,ue_ap_1,ue_ap_2,ue_ap_3,ue_ap_4
end goal

environment()

```

FIGURE 6. Role of session, environment, and goal.

<pre> % OFMC % Version of 2006/02/13 SUMMARY SAFE DETAILS BOUNDED_NUMBER_OF_SESSIONS PROTOCOL /home/span/span/testsuite/results/tenth2.if GOAL as_specified BACKEND OFMC COMMENTS STATISTICS parseTime: 0.00s searchTime: 0.64s visitedNodes: 64 nodes depth: 6 plies </pre>	<pre> SUMMARY SAFE DETAILS BOUNDED_NUMBER_OF_SESSIONS TYPED_MODEL PROTOCOL /home/span/span/testsuite/results/tenth2.if GOAL As Specified BACKEND CL-AtSe STATISTICS Analysed : 13699 states Reachable : 13699 states Translation: 0.06 seconds Computation: 31.42 seconds </pre>
--	--

FIGURE 7. Simulation results of AVISPA simulation.

VI. PERFORMANCE ANALYSIS

In this section, we compare the computation, communication costs, and security features of the proposed scheme with

existing schemes [13], [14], [15], [16], [17], [18]. These schemes are proposed in the 6G-enabled networks, which are similar to the those of proposed scheme. The existing schemes [13], [14], [15], [16], [17], [18] assumed that the channel between AP and CN was secure. Therefore, some papers may have lower computational costs than the proposed method, yet the proposed scheme can provide superior security than existing schemes.

TABLE 5. Notations, meaning operations, and time costs.

Notation	Meaning
T_{mul}	Time cost of ECC scalar multiplication
T_{add}	Time cost of ECC point addition
T_h	Time cost of hash operation
T_r	Time cost of random number generation
T_{sym}	Time cost of symmetric key en/decryption

A. COMPUTATIONAL COST

We compare the computational cost of our scheme and existing schemes. We refer to the time cost of each operation as measured in [13]. Darman et al. measure time costs for mobile devices, servers, and drones, respectively. They used the iPhone Xs Max with 8GB RAM, 1.6 GHz Quad-Core Processor, and 15.1 IOS operating system representing mobile devices, a Dell Ultrabook 8757P with 8GB RAM and Intel I5-6300C representing servers, and a Raspberry PI 3 B+ with a Cortex-A53 64-bit representing drones. In our scheme, these devices can represent user equipment, core network, and access point, respectively. Referring to the experimental results in [13], we compare computational costs by adding up the time costs of all operations used in each authentication protocol. The notation of each operation are summarized in Table 5 and the time costs of operations in each environment are summarized in Table 6.

TABLE 6. Time costs of operations in each environment.

Notation	UE	AP	CN
T_{mul}	5.116 ms	4.107 ms	0.926 ms
T_{add}	0.018 ms	0.013 ms	0.006 ms
T_h	0.009 ms	0.006 ms	0.004 ms
T_r	2.011 ms	1.185 ms	0.118 ms
T_{sym}	0.017 ms	0.013 ms	0.008 ms

Based on the results presented in Table 6, the computational cost of the proposed scheme and existing schemes are summarized in Table 7. Consequently, the proposed scheme incurs more computational costs compared to [14], [16], [17], and [18]. It is because these schemes are designed assuming an existence of secure channels or do not consider various attack vectors such as ESL attack, and these schemes can have security problems in real environments. Despite being designed without assuming a secure channel, the proposed scheme incurs comparable computations costs compared to existing schemes [13], [15]. It means that the proposed scheme can guarantee usability for users when the proposed scheme applied to a real environment because it has lower network delay caused by computational costs compared to existing schemes.

TABLE 7. Computational cost comparison.

Scheme	UE_k	AP_j	CN_i
Darman et al. [13]	$6T_{mul} + T_{add} + 19T_h + T_r \approx 32.896ms$	$5T_{mul} + T_{add} + 9T_h + T_r + 2T_{sym} \approx 21.813ms$	$5T_{mul} + T_{add} + 12T_h + T_r \approx 4.802ms$
Chaudhry et al. [14]	$3T_h + T_r + T_{sym} \approx 2.055ms$	$7T_h + T_r + 5T_{sym} \approx 1.292ms$	$3T_h + T_r + T_{sym} \approx 1.216ms$
Ma et al. [15]	$4T_{mul} + 7T_h + 2T_r + 2T_{sym} \approx 24.583ms$	—	$4T_{mul} + 7T_h + 2T_r + 2T_{sym} \approx 3.984ms$
Zhang et al. [16]	$3T_{mul} + 7T_h + T_r + T_{sym} \approx 17.439ms$	$2T_{mul} + 6T_h + T_r + 2T_{sym} \approx 9.461ms$	$T_{mul} + 9T_h + 3T_{sym} \approx 0.986ms$
Deebak et al. [17]	$3T_{mul} + 6T_h + T_r \approx 17.413ms$	$3T_{mul} + 6T_h + T_r \approx 13.542ms$	$8T_h \approx 0.032ms$
Roy et al. [18]	$2T_{mul} + 3T_h + 2T_r + 2T_{sym} \approx 14.315ms$	$2T_{mul} + 2T_h + T_{sym} \approx 8.239ms$	—
Proposed AP-CN	—	$4T_{mul} + 3T_h + T_r \approx 17.631ms$	$4T_{mul} + 3T_h + T_r \approx 3.834ms$
Proposed UE-AP	$4T_{mul} + 4T_h + T_r \approx 22.511ms$	$4T_{mul} + 3T_h + T_r \approx 17.631ms$	—

— : No computation cost.

B. COMMUNICATION COST

We also analyzed the communication cost of the proposed scheme and existing schemes [13], [14], [15], [16], [17], [18]. We assume that the bit length of a hash output, an ECC point, a random number, an identity, and a timestamp are 256 bits, 320 bits, 192 bits, 192 bits, and 32 bits, respectively. Then, the communication cost generated in each entity are as summarized in Table 9.

In terms of communication cost, the proposed scheme generates lower cost than existing schemes most of [13], [14], [15], and [16]. Although the proposed scheme has slightly higher communication costs than [17], [18], the proposed scheme can provide better security as we demonstrate in Section VI-C.

C. SECURITY FEATURES

We compared the security features of the proposed scheme with those of related schemes [13], [14], [15], [16], [17], [18]. We considered A1) “resistance to replay and MITM attacks,” A2) “resistance to device stolen attack,” A3) “access point capture attack,” A4) “resistance to impersonation attack,” A5) “resistance to session key disclosure attack,” A6) “preservation of anonymity and untraceability,” A7) “support access control,” A8) “not assuming an existence of a secure channel,” and A9) “resistance to ESL attack”. Table 8 summarizes the security features of the proposed scheme and related schemes. Table 8 shows that the proposed scheme can provide more security features than related schemes in similar environments [13], [14], [15], [16], [17], [18]. In summary, the proposed scheme may incur more computational costs than some existing schemes in the similar environments. However, the proposed scheme is more efficient than the existing schemes in terms of communication costs and is much better in terms of security because the proposed scheme considers zero-trust model for 6G environments. Therefore, it is clear that the proposed scheme can be practical in 6G-enabled IoT environments

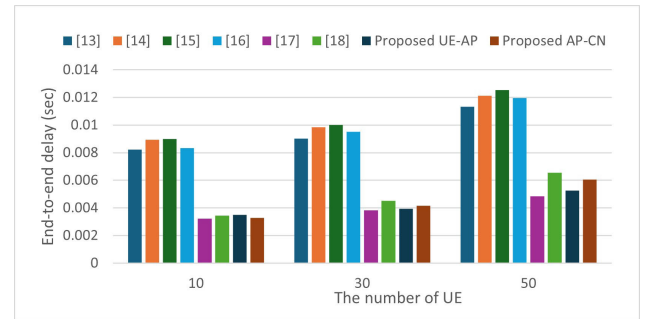
D. EXPERIMENTAL ANALYSIS USING NS-3

We experiment the proposed scheme using network simulator 3 (NS-3) [39] to evaluate the practicality our scheme and other schemes. For the proposed two authentication

TABLE 8. Comparison of security features.

Features	[13]	[14]	[15]	[16]	[17]	[18]	Proposed
A1	O	O	O	O	O	O	O
A2	O	—	O	O	O	O	O
A3	O	O	O	—	—	X	O
A4	O	O	O	O	O	O	O
A5	O	O	O	O	O	O	O
A6	O	O	O	O	O	O	O
A7	—	—	—	—	—	—	O
A8	X	X	X	O	X	X	O
A9	—	O	O	O	—	X	O

X: Insecure, O: Secure, —: Not considered.

**FIGURE 8. Measured end-to-end delay of the proposed scheme and related schemes.**

schemes, we measured throughput and end-to-end delay with increasing the number of user equipment. In the proposed AP-CN authentication, we consider AP as UE for the simulation. Table 10 shows the simulation parameters and environments.

The end-to-end delay is calculated as $\sum_{i=1}^{n_p} (T_p - T_s) / n_p$. i , T_p represents a data packet, T_s represents a time cost for receiving and sending messages, and n_p represent the total number of packets. We can obtain the average time it takes to send and receive packets by measuring the end-to-end delay. The throughput can be formulated as $n_{rp} * |s| / T_s$. n_{rp} is the number of received packets, $|s|$ is the packet size, and T_s is the total time in second. Figure 8 and 9 summarize the comparison of simulation results of the proposed scheme and related schemes.

The both end-to-end delay and throughput depends on the communication cost. Accordingly, The schemes of [13], [14], [15], and [16] have high throughput and end-to-end delay, and it means that users can experience a delay while sending and

TABLE 9. Communication cost comparison.

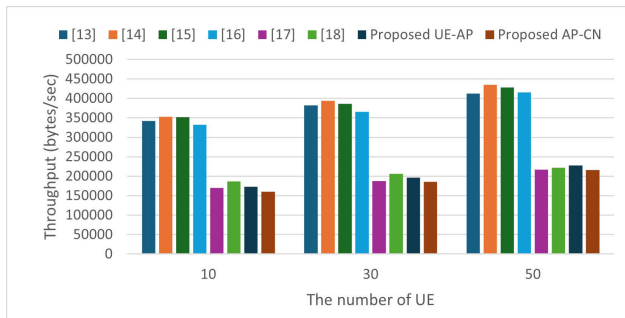
Scheme	UE_k	AP_j	CN_i	Total
Darman <i>et al.</i> [13]	1568 bits	1600 bits	1408 bits	4576 bits
Chaudhry <i>et al.</i> [14]	928 bits	2272 bits	1728 bits	4928 bits
Ma <i>et al.</i> [15]	1568 bits	1824 bits	1568 bits	4960 bits
Zhang <i>et al.</i> [16]	1344 bits	2464 bits	800 bits	4608 bits
Deebak <i>et al.</i> [17]	800 bits	908 bits	—	1708 bits
Roy <i>et al.</i> [18]	832 bits	1088 bits	—	1920 bits
Proposed AP-CN	—	800 bits	800 bits	1600 bits
Proposed UE-AP	1120 bits	800 bits	—	1920 bits

— : No communication cost.

TABLE 10. Simulation parameters.

Parameters	Values
Operating systems	Ubuntu 16.04 LTS
Simulator	NS 3.29
Simulation area	1000 * 1000m ²
Mobility model	RandomWalk2dMobilityModel
Propagation loss model	LogDistancePropagationLossModel
Channel Width	160 MHz
Channel type	Wireless
Routing protocol	AODV
Simulation time	10 minutes

receiving messages. The proposed scheme and the schemes of [17] and [18] have relatively lower throughput and end-to-end delay, and it means that users can send and receive messages faster than the those of other schemes.

**FIGURE 9. Measured throughput of the proposed scheme and related schemes.**

VII. CONCLUSION

This paper proposed zero-trust AP-CN and UE-AP authentication schemes and a fine-grained access control scheme for 6G-enabled IoT environments. Many authentication protocols have been proposed for 5G and 6G networks. However, existing schemes have not considered the zero-trust model and designed authentication protocols assuming the existence of a secure channel. These schemes cannot satisfy the complex security requirements in 6G communication environments. We developed blockchain-based authentication schemes, and the proposed scheme can guarantee zero-trust security by separately designing an authentication protocol between the AP and CN. We analyzed the proposed scheme formally and informally and compared the computational and communication costs with those of the existing schemes to demonstrate the superiority of the proposed scheme. The proposed scheme introduced the

zero-trust-based system model for 6G-enabled IoT networks using blockchain. We believe that the proposed scheme can be adopted in 6G networks with aerial base station or mobile control center, as these networks are decentralized and there are no secure channels in the network. Consequently, the proposed scheme can be practically applied in 6G-enabled communication environments. In future work, we aim to reduce the computational cost of the proposed protocol to enhance to efficiency of our scheme.

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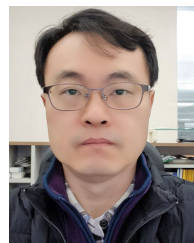
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